

Agentic Tutor MVP for Chinese A-Level Students

Executive summary

A strong MVP for an “agentic tutor” aimed at Chinese Cambridge A-Level students in Chemistry 9701, Physics 9702, Economics 9708, and Mathematics 9709 should be a **tool-routed web app**, not a heavy autonomous multi-agent system. The product needs to handle four very different assessment patterns: Chemistry and Physics include multiple-choice, structured responses, and practical-skills-oriented assessment; Economics includes multiple choice, data response, and essays; Mathematics is dominated by structured written work across pure mathematics, mechanics, and statistics. That means the MVP should optimize for **image upload, question parsing, shown-working grading, bilingual tutoring, and cited explanations**, rather than for broad web autonomy. ¹

The most practical low-cost implementation is a **China-region web stack** with **Next.js frontend**, **FastAPI backend**, **PostgreSQL + Redis + object storage**, and **hosted open-source model APIs** as the intelligence layer. For the model layer, the cleanest default is **Alibaba Cloud Model Studio 百炼** because it offers China-mainland deployment, OpenAI-compatible APIs, Qwen open-source model families, OCR, multimodal models, batch inference, and context caching in one platform. The most budget-efficient core combination is: **Qwen2.5-7B or Qwen3-8B** for normal tutoring, **Qwen2.5-Math-7B** for math-heavy grading, **Qwen-VL-OCR** for cheap OCR, and **Qwen3-VL-8B** as a vision fallback on hard diagrams/graphs/images. Tencent OCR should be treated as a **pilot-time helper** for question splitting and formula recognition because its free resource packs are helpful, but its paid question-grading path becomes expensive quickly under a ¥20/month API ceiling. ²

Pedagogically, the app should be built around **explicit modeling, worked examples, metacognitive prompting, and actionable feedback**. In practice, that means every answer should follow a stable teaching shape: **simple restatement → clever hint → step plan → full worked answer → why this earns marks → citations**. This is much closer to what evidence-backed tutoring and teacher feedback guidance recommends than either “just chat” or dumping full solutions immediately. ³

Under the stated constraint of **monthly API budget ≤ ¥20 CNY**, the MVP is feasible only if you assume **private-alpha traffic**, short prompts, compressed images, aggressive caching, and “full grading” only when the student asks for it. That budget is realistic for a founder pilot, a single tutor, or a small school trial. It is **not** realistic for an open consumer launch unless you add subscription revenue, BYOK, school billing, or a higher budget envelope. The good news is that the underlying token prices are low enough that a narrowly-scoped tutor remains viable: for example, Alibaba’s China-mainland pricing lists **Qwen-VL-OCR at ¥0.3/M input tokens and ¥0.5/M output tokens**, **Qwen2.5-7B-Instruct at ¥0.5/M input and ¥1/M output**, **Qwen2.5-Math-7B-Instruct at ¥1/M input and ¥2/M output**, and **Qwen3-VL-8B-Instruct at ¥0.5/M input and ¥2/M output**. ⁴

Assumptions and design principles

I am making the following assumptions because they were not specified. The MVP is assumed to be a **private beta** with roughly **20–80 registered students**, mostly **ages 15–19**, with **300–1,500 tutoring turns per month** and **100–400 image uploads per month**. I also assume the **¥20 cap applies to model and OCR APIs only**, not to infrastructure, domain, or SMS costs. Finally, I assume the product

will initially rely on **user-uploaded questions, teacher-authored notes, and syllabus-linked concept notes**, rather than redistributing a large built-in Cambridge past-paper archive. These are assumptions, not sourced facts.

The first design principle should be **exam realism**. The tutor must reflect how these subjects are actually assessed, because the paper formats strongly influence both UI and grading logic. Chemistry and Physics include structured questions plus practical-skills-oriented assessment. Economics uses data response and essay writing. Mathematics focuses on structured written work and explicitly expects students to show their working on the paper. That is why the app should not only answer questions; it must also judge method, notation, units, structure, and exam communication. ⁵

The second principle should be **agentic, but narrow**. For an MVP with a tiny budget, “agentic” should mean **structured tool routing** rather than “many autonomous agents debating each other”. One orchestrator should classify the request, then call the cheapest appropriate tool chain: plain text tutor, OCR pipeline, multimodal parser, grader, citation retriever, or homework batch job. This architecture reduces tokens, makes costs predictable, and is easier to test. Alibaba’s OpenAI-compatible API surface, batch inference, and context caching make this much easier to implement cleanly. ⁶

The third principle should be **bilingual teaching with exam-English preservation**. Since the target learners are Chinese students but the qualification is Cambridge A-Level, the product should allow Chinese scaffolding without hiding the English exam language. The right compromise is not full translation of everything. It is: keep subject terms, formulas, command verbs, and exam-style final answers in English; offer Chinese glosses, metaphors, and rephrasings around them; let users choose **English-first, Chinese-first, or Exam English mode**. This is a product recommendation, but it is also consistent with research-backed emphasis on explicit modeling, gradual release, and clear feedback. ⁷

The fourth principle should be **citation-backed explainability**. Students should not receive “because the AI says so.” They should see what the answer is grounded in: syllabus topic references, formula-sheet references, glossary entries, and licensed or school-authored concept notes. Cambridge officially makes syllabuses and subject resources discoverable through its subject pages, and Cambridge also states that teachers at Cambridge schools may use past papers in class, but those materials should not be republished or resold without approval. For an MVP, the safest route is to cite **syllabus snippets and your own concept bank**, and to treat licensed past papers/mark schemes as restricted, school-side assets rather than a public content layer. ⁸

The fifth principle should be **data minimization from the beginning**. Under China’s Personal Information Protection Law, sensitive personal information carries higher obligations, and information about under-14 minors is specifically treated as sensitive with special handling expectations. The CAC’s rules on generative AI services also apply where such services are offered to the public in China, and cross-border transfer creates an additional compliance layer. In practice, the MVP should therefore default to **China-region storage and model endpoints**, collect minimal profile data, support deletion/export, and separate “learning telemetry” from personally identifying fields. ⁹

MVP scope and learning experience

Product scope

The MVP should focus on a small, coherent set of capabilities that directly improve learning and can be evaluated quickly.

Area	Must ship in MVP	Record now, surface later	Defer beyond MVP
Question solving	Text input, image upload, hint-first tutoring, full worked answer	Per-topic mastery trends	Autonomous practice-plan generation
Grading	Short-answer grading, method checking, unit/notation checks, essay feedback skeleton	Wrong-question notebook entries	High-stakes final scores for long essays
OCR / vision	Printed text OCR, formula extraction, basic graph/diagram understanding, image crop QA	Region-level OCR confidence heatmap	Robust grading of messy freehand organic mechanisms
Bilingual UX	English-first / Chinese-first / Exam-English modes	Per-user language preference and switching behavior	Full voice tutoring
Explainability	Citation chips to syllabus/concept notes, mark breakdown, "why this is wrong"	Confidence-based review flags	Open-ended "AI opinions" with no grounding
Homework	Upload worksheet/homework page, split into questions, batch correction queue	Homework trend analytics	Teacher workflow suite

This scope matches the assessment demands of the four target subjects. Chemistry and Physics need structured and practical-style reasoning; Economics needs data-response and essay scaffolding; Mathematics needs shown-working and component-aware feedback. ⁵

Bilingual UX model

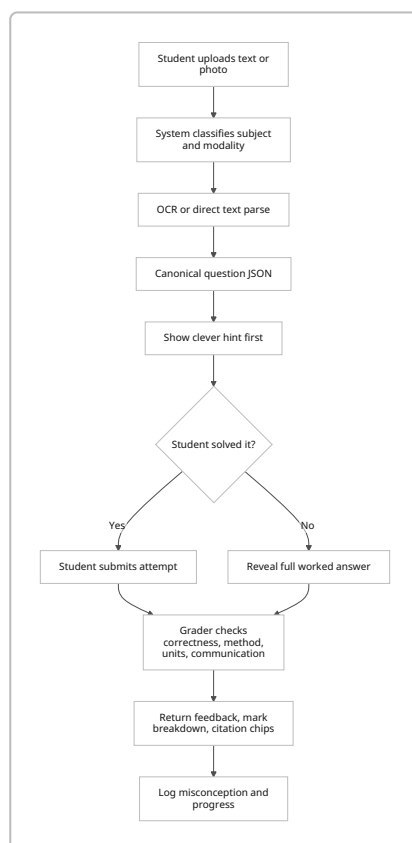
The UX should be intentionally simple. On every question page, the student should see only four primary controls above the answer area: **Language mode**, **Need a hint**, **Show full answer**, and **Grade my attempt**. Everything else can stay collapsed.

Mode	Main response language	What stays in English	Best use case
English-first	English	All terms, formulas, final answer	Stronger students, exam prep
Chinese-first	Chinese	Keywords, formulas, final answer line, command words	First-pass understanding
Exam English	English only	Everything	Timed practice and revision
Hover gloss	Same as main mode	English terms show Chinese tooltip	Vocabulary building

The most important UX rule is this: **the student should always understand what to do next**. That means every reply begins with a single clear line such as: “Try this first,” “Check this sign,” or “The key idea is that pressure pushes equilibrium toward fewer gas molecules.” This is where the app becomes tutor-like rather than chatbot-like, and it aligns well with evidence favoring explicit modeling and actionable feedback. 10

Core user flows

A single-question flow should feel like this:



This flow is deliberately staged because the tutor should not erase the student’s productive struggle too early. The “clever hint first” stage is a pedagogical feature, not only a UX feature. It supports metacognitive planning before the student sees the worked solution. 7

A homework-correction flow should be asynchronous. The student uploads one or more images, the system preprocesses and splits them into questions, then returns a queue view showing **Ready / Needs better image / Low confidence / Reviewed**. This matters because Chemistry, Physics, and Mathematics often involve graphs, tables, formulas, multi-part working, and diagrams, while Economics homework can mix diagrams, short analysis, and longer prose on the same page. 5

Subject-specific teaching behaviors

Subject	Tutor emphasis	High-value UI extras
Chemistry 9701	Mole ratios, equilibrium logic, organic transformations, analysis, practical interpretation	Inline formula rendering, reaction-scheme canvas, common ion/mechanism warnings
Physics 9702	Units, sign conventions, free-body diagrams, graph reading, energy logic, practical/data reasoning	Unit checker, sketch-pad overlay, graph annotation
Economics 9708	Definitions, causal chains, diagrams, evaluation balance, essay structure	"Because... therefore..." sentence stems, evaluation checklist
Mathematics 9709	Shown working, notation normalization, algebraic method, mechanics/statistics structure	Step validator, LaTeX rendering, "final answer box"

This subject split follows the actual syllabus/assessment structures rather than generic "STEM vs humanities" grouping. ¹¹

Sample metaphors and explanations

The app should maintain a small bank of reusable metaphors. The goal is not cuteness. The goal is **recall**.

Subject	Sample aligned topic	Memorable metaphor	Clever hint first	Full-answer style
Chemistry 9701	Equilibrium and pressure	Crowded lift: when the lift gets crowded, equilibrium prefers the side with fewer "people" (gas particles).	"If pressure rises, ask: which side has fewer gas particles?"	"Increasing pressure shifts the equilibrium to the side with fewer moles of gas. In $\text{N}_2 + 3\text{H}_2 \rightleftharpoons 2\text{NH}_3$, the left has 4 moles of gas and the right has 2, so the equilibrium shifts right and ammonia yield increases."
Physics 9702	Work, energy and power	Height coins turning into speed coins	"Before using SUVAT, ask whether energy changes more simply."	"Ignoring air resistance, gravitational potential energy becomes kinetic energy: $mgh = \frac{1}{2}mv^2$. So $v = \sqrt{2gh}$. The mass cancels, which is a good check."

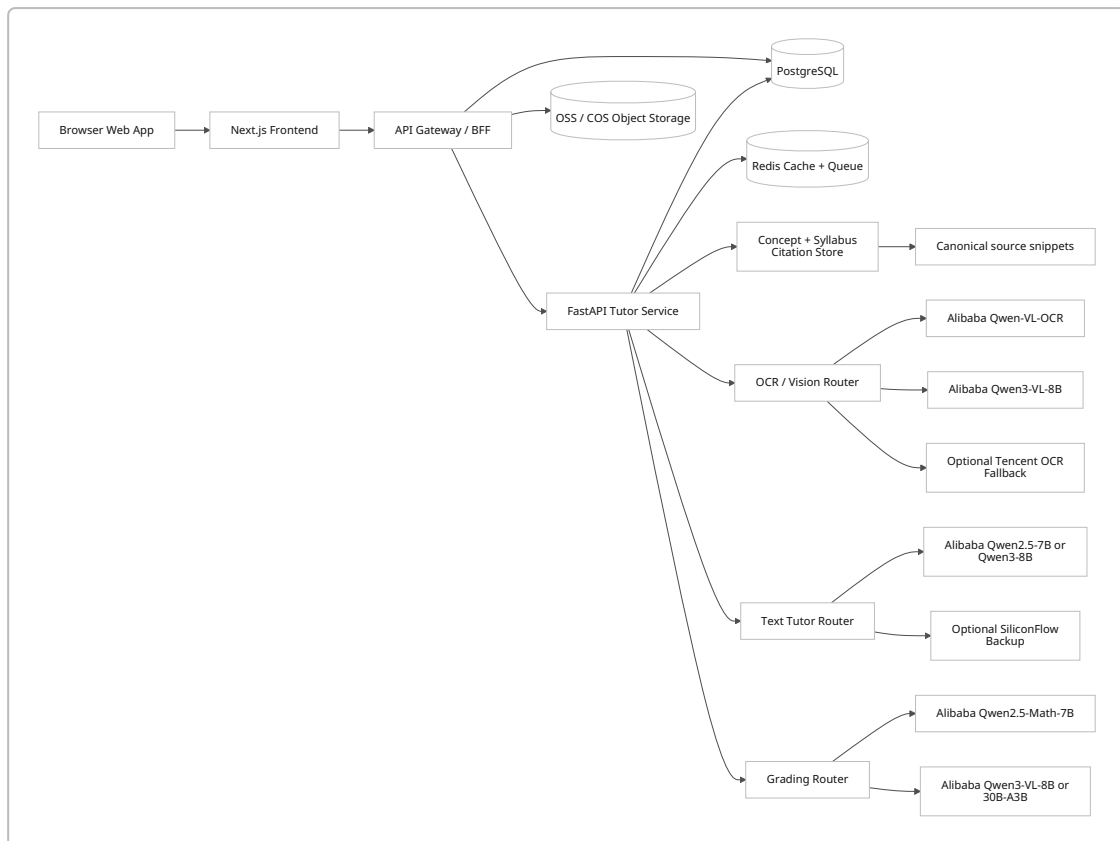
Subject	Sample aligned topic	Memorable metaphor	Clever hint first	Full-answer style
Economics 9708	Exchange rates / current account	Shop sale sticker: depreciation puts a sale sticker on exports and a higher price tag on imports.	"Think in two doors: foreign buyers and domestic buyers."	"A depreciation tends to make exports cheaper for foreigners and imports dearer for domestic consumers. If quantities respond enough, export spending rises and import spending falls, improving the current account."
Mathematics 9709	Mechanics on an incline	Gravity split into straight-down and slide-along pieces	"Do not resolve the normal reaction. Resolve the weight."	"On a smooth plane inclined at 30° , the downslope component is $mg \sin 30^\circ$. Hence $a = g \sin 30^\circ = 4.9 \text{ m s}^{-2}$."

These sample topics are all inside the official subject content areas of the four syllabuses. 12

System architecture and implementation design

Reference architecture

The recommended MVP architecture is a **web BFF + AI service** pattern. It keeps the UI fast, the auth simple, and the model orchestration testable.



Alibaba’s official documentation is especially helpful here because it gives you China-mainland deployment controls, OpenAI-compatible APIs, batch inference, context caching, OCR, multimodal models, and open-model pricing in one place. That makes it easier to keep both data residency and cost under control. ¹³

Why this architecture fits the budget

The budget constraint is so tight that the architecture must avoid waste. Three design choices matter most.

First, use a **cheap-text-first router**. Most tutoring turns are not vision-heavy. If a question arrives as text, or if OCR confidence is already high, route directly to a low-cost open model such as **Qwen2.5-7B-Instruct** or **Qwen3-8B** instead of calling a larger multimodal model. Alibaba’s China-mainland pricing for these smaller open Qwen models is far lower than premium reasoning routes. ¹⁴

Second, treat OCR and multimodal analysis as **separate stages**. Use **Qwen-VL-OCR** to extract text cheaply; call **Qwen3-VL-8B** only when OCR is low confidence, the question includes a graph/diagram/table, or the student’s answer itself is visual. Since Alibaba prices OCR and VLM separately, this split keeps routine workflows extremely cheap. ¹⁵

Third, cache aggressively. Alibaba documents both implicit and explicit context caching, and explicitly notes that stable prompt reuse can materially lower cost; it also offers batch inference at half price for non-real-time jobs. The practical implication is simple: keep system prompts and syllabus instructions stable, cache them, and run nightly jobs such as exercise summarization, notebook generation, and analytics asynchronously. ¹⁶

Internal module interfaces

A clean internal contract matters more than microservice purity at MVP stage.

Module	Responsibility	Key input	Key output
QuestionClassifier	Detect subject, modality, answer type, required tools	raw text/ image metadata	subject, modality, question_type, confidence
OcrPipeline	Preprocess image, OCR, formulas, regions, confidence scoring	image/PDF page	CanonicalQuestionSet
QuestionNormalizer	Clean OCR text, attach marks/ subparts/diagram refs, render formulas to LaTeX	OCR raw output	CanonicalQuestion
TutorEngine	Produce hint-first and full-answer responses	canonical question + context	HintResponse, SolutionResponse
GraderEngine	Compare student attempt with target method and rubric	canonical question + student answer	GradeResult
CitationEngine	Retrieve syllabus/ concept snippets and attach source chips	subject/topic tags	SourceSnippet[]
ErrorLogger	Persist misconception tags, confidence gaps, retry counts	grade result + usage events	ErrorLog
HomeworkBatchRunner	Queue entire worksheets and aggregate results	assignment upload	HomeworkReport

A good canonical question schema is the hinge that keeps the rest of the system stable:

```
{
  "question_id": "q_9709_2026_demo_001",
  "subject": "9709",
  "question_type": "structured_calculation",
  "language_mode": "en_first",
  "stem_text": "A block rests on a smooth plane inclined at 30°...",
  "subparts": [
```



```

    { "label": "(a)", "marks": 2, "target_skill": "resolve_forces" }
  ],
  "diagram_regions": [
    { "bbox": [120, 88, 420, 360], "kind": "diagram" }
  ],
  "formula_regions": [
    { "bbox": [460, 102, 700, 220], "latex": "F = ma" }
  ],
  "ocr_confidence": 0.93,
  "source_refs": ["9709_mechanics_component"],
  "needs_visual_reasoning": false
}

```

External API endpoints

A minimal HTTP surface is enough for MVP.

Endpoint	Method	Purpose	Notes
/auth/register	POST	Email/password or email-OTP registration	Keep SMS out of MVP
/uploads/presign	POST	Signed upload URL for image/PDF	Store originals in object storage
/questions/parse	POST	OCR + normalization + classification	Async for large uploads
/tutor/hint	POST	Return hint-first answer	Fast path, short output cap
/tutor/solve	POST	Return full worked explanation	Called only on demand
/grading/grade	POST	Grade student attempt	Supports text and image answers
/homework/submit	POST	Create homework batch job	Returns <code>job_id</code>
/homework/:job_id	GET	Job status and results	Poll or use SSE
/progress/summary	GET	Topic mastery and recent mistakes	MVP dashboard
/errors/log	POST	Record low-level telemetry	No UI required yet

OCR and homework-correction pipeline

The OCR pipeline should be deliberately hybrid, because no single low-cost route is best for all cases.

The cheapest robust path is: **client-side image compression → server-side deskew/crop with classic CV → cheap OCR → confidence check → vision fallback only if needed**. This works because “no local models” does not prevent you from using deterministic local image operations such as resize, thresholding, contour detection, or perspective correction. The expensive part is model inference, so classical preprocessing is worth doing locally on the server. The OCR/model stages are grounded in the

capabilities exposed by Alibaba's OCR/VLM stack and Tencent's question/formula OCR: Qwen-OCR supports text extraction and structured parsing; Qwen3-VL is positioned for image/question understanding; Tencent's question recognition and formula/math OCR support coordinates, handwriting, and LaTeX outputs. ¹⁷

A good homework correction pipeline is:

1. **Upload** page image(s) or a PDF.
2. **Client validation**: reject tiny, blurry, rotated, or overexposed images.
3. **Preprocess** with OpenCV: crop borders, deskew, normalize contrast, compress longest edge to roughly 1280–1600 px.
4. **Question split**:
 5. cheapest default: contour/layout heuristics + subpart markers;
 6. optional pilot-time fallback: Tencent `试卷切题` if heuristics fail.
7. **Primary OCR** with `qwen-v1-ocr`.
8. **Formula recovery**:
 9. if OCR text contains math/chem syntax loss, call Tencent formula/math OCR during pilot, or use VLM fallback;
10. normalize formulas to LaTeX.
11. **Diagram/graph pass** with `qwen3-v1-8b-instruct` when question includes axes, curves, apparatus, diagrams, or image confidence is low.
12. **Question normalization** into canonical JSON.
13. **Answer-type-specific grading**:
 14. exact/MCQ,
 15. structured calculation,
 16. short explanation,
 17. data response,
 18. essay skeleton feedback.
19. **Feedback assembly** with hint status, full solution, marks, citations, and misconception tags.
20. **Store telemetry** for future wrong-question notebook and profiling.

The key product decision is that **image understanding and grading are not the same stage**. First recover what the question and answer are. Then grade the meaning. That separation gives better debugging, better explainability, and lower cost.

Explainability and citation design

The app should have a small **citation store** rather than letting the model invent references. Each answer generator call should receive retrieved snippets from:

- official Cambridge syllabus sections,
- formula sheets where relevant,

- teacher-authored concept notes,
- school-owned glossary entries.

The answer renderer should then force a stable shape:

```
Core idea
Clever hint
Plan
Worked answer
Why this earns marks
Common trap
Sources
```

The model should only cite sources from the retrieved set, never arbitrary web text. This avoids hallucinated references and makes answers easy to audit.

A practical citation format for the UI would be compact chips such as:

- 9701 \$Equilibria
- 9702 \$Work, energy and power
- School Note PH-12
- Formula Sheet M1

This is a better explainability standard than surfacing raw model chain-of-thought. The student sees the teaching logic and the evidence, not internal hidden reasoning.

Prompt templates

The prompt templates below are intentionally short. Under a ¥20 monthly limit, prompt discipline is part of product design.

Tutor hint prompt

```
System:
You are an A-Level tutor for Cambridge subjects 9701/9702/9708/9709.
Teach in {language_mode}. Use simple sentences. Keep exam terms in English.
Give only:
1) one-line core idea
2) one clever hint
3) one "what to do next" step
Do NOT reveal the full solution unless asked.
If uncertain from OCR, say what is unclear.

User:
Subject: {subject}
Question type: {question_type}
Question:
{question_text}
Student attempt:
{student_attempt_or_blank}
```

Sources:
{retrieved_snippets}

Full solution prompt

System:
You are an exam-focused A-Level tutor.
Teach in {language_mode}; preserve formulas and command words in English.
Output exactly in this order:
Core idea
Metaphor
Step plan
Worked solution
Final exam-style answer
Why this gets marks
Common mistake
Sources

Rules:

- Simple language
- Show calculations clearly
- Prefer one memorable metaphor
- Cite only from provided sources
- If the image is ambiguous, mark low-confidence

User:
{canonical_question_json}
{retrieved_snippets}

Grading prompt

System:
You are a cautious grader, not a generous chatbot.
Score according to the rubric and explain point-by-point.
If handwriting/OCR is uncertain, lower confidence instead of guessing.

Rubric:

- Correctness
- Method / reasoning
- Units / notation / diagram labels
- Exam communication

Output JSON:

```
{
  "score_awarded": 0,
  "score_max": 0,
  "confidence": 0.0,
  "verdict": "",
  "strengths": [],
```

```

    "mistakes": [],
    "misconception_tags": [],
    "next_step": "",
    "citations": []
  }

```

User:

Question: {canonical_question}

Mark points / expected method: {marking_template}

Student answer: {student_answer}

Sources: {retrieved_snippets}

OCR cleanup prompt

System:

You are a correction layer for OCR output from A-Level questions.
Repair broken line breaks, merge split equations, keep all symbols, preserve English terms,
and return structured JSON only.

User:

Subject: {subject_hint}

Raw OCR:

{raw_ocr_text}

Detected regions:

{bbox_list}

Answer and grading templates

A stable answer template is better than free-form prose.

Question type	Output template
MCQ	Core idea → elimination hint → correct option → one-line reason
Structured calculation	Core idea → formula/plan → line-by-line working → boxed final answer → mark points
Short explanation	Core idea → metaphor → two to four reasoning steps → concise exam answer
Data response	Read the data → identify trend → explain with economics concept → evaluate caveat
Essay	Definitions → diagram/structure → argument chain → evaluation → conclusion
Practical / graph / diagram	What is shown → what matters → inference from data/shape → common mistakes

A simple MVP grading rubric can work across subjects if the weighting changes by type.

Dimension	Default weight	Notes
Correctness	40%	Final answer or conclusion
Method / reasoning	35%	Most important in Maths, Physics, Chemistry
Representation	15%	Units, notation, graph labels, diagrams
Communication	10%	Especially important in Economics prose

Data model and telemetry

The data model should be optimized for three things: **auditability**, **incremental explainability**, and **future profiling**. Because the target subjects use structured papers, diagrams, multi-part questions, and shown working, the schema should preserve intermediate artifacts such as OCR regions, normalized formulas, citations, and error tags. ¹⁸

Questions table

Column	Type	Notes
<code>id</code>	UUID PK	Internal question id
<code>user_id</code>	UUID FK nullable	Null for shared seeded/demo items
<code>subject_code</code>	VARCHAR(4)	<code>9701</code> , <code>9702</code> , <code>9708</code> , <code>9709</code>
<code>question_type</code>	TEXT	<code>mcq</code> , <code>structured_calc</code> , <code>essay</code> , <code>data_response</code> , <code>practical</code> , <code>graph</code> , <code>diagram</code>
<code>source_type</code>	TEXT	<code>upload</code> , <code>manual</code> , <code>teacher_bank</code> , <code>demo</code>
<code>raw_text</code>	TEXT	Original text or OCR output
<code>normalized_text</code>	TEXT	Cleaned stem
<code>canonical_json</code>	JSONB	Full structured representation
<code>marks_total</code>	INT nullable	If known
<code>command_words</code>	TEXT[]	e.g. <code>calculate</code> , <code>explain</code> , <code>evaluate</code>
<code>has_formula</code>	BOOL	Parsed formula present
<code>has_diagram</code>	BOOL	Diagram region present
<code>has_graph</code>	BOOL	Graph/axes region present
<code>ocr_confidence</code>	NUMERIC(4,3) nullable	Aggregate confidence
<code>parse_confidence</code>	NUMERIC(4,3)	Post-normalization confidence
<code>created_at</code>	TIMESTAMP	
<code>updated_at</code>	TIMESTAMP	

Recommended indexes: `(user_id, subject_code, created_at DESC)` and a GIN index on `canonical_json`.

Answers table

Column	Type	Notes
<code>id</code>	UUID PK	
<code>question_id</code>	UUID FK	
<code>user_id</code>	UUID FK	
<code>answer_mode</code>	TEXT	<code>hint</code> , <code>full_solution</code> , <code>grading_result</code> , <code>homework_batch_item</code>
<code>language_mode</code>	TEXT	<code>en_first</code> , <code>zh_first</code> , <code>exam_en</code>
<code>student_input_text</code>	TEXT nullable	Student typed attempt
<code>student_input_asset_id</code>	UUID nullable	Image answer upload
<code>hint_text</code>	TEXT nullable	First-stage response
<code>full_solution_text</code>	TEXT nullable	Full answer
<code>score_awarded</code>	NUMERIC(5,2) nullable	
<code>score_max</code>	NUMERIC(5,2) nullable	
<code>grader_confidence</code>	NUMERIC(4,3) nullable	
<code>rubric_json</code>	JSONB	Dimension-level scores
<code>citations_json</code>	JSONB	Source refs rendered in UI
<code>llm_model</code>	TEXT	Exact model id used
<code>provider</code>	TEXT	<code>bailian</code> , <code>tencent</code> , <code>siliconflow</code>
<code>token_usage_json</code>	JSONB	Per-call cost audit
<code>created_at</code>	TIMESTAMP	

Annotations table

Column	Type	Notes
<code>id</code>	UUID PK	
<code>question_id</code>	UUID FK	
<code>asset_id</code>	UUID FK	Linked image/PDF page

Column	Type	Notes
<code>region_type</code>	TEXT	<code>question_box</code> , <code>subpart</code> , <code>formula</code> , <code>graph</code> , <code>diagram</code> , <code>student_work</code> , <code>citation_highlight</code>
<code>bbox_json</code>	JSONB	Bounding box coordinates
<code>text_value</code>	TEXT nullable	OCR text for region
<code>latex_value</code>	TEXT nullable	Formula output
<code>label</code>	TEXT nullable	e.g. <code>(a)</code> , <code>axis_x</code> , <code>wrong_sign_here</code>
<code>confidence</code>	NUMERIC(4,3)	Detection confidence
<code>review_status</code>	TEXT	<code>auto</code> , <code>flagged</code> , <code>approved</code>
<code>created_at</code>	TIMESTAMP	

User progress table

Column	Type	Notes
<code>user_id</code>	UUID PK/FK	
<code>preferred_language_mode</code>	TEXT	
<code>subject_focus_json</code>	JSONB	e.g. weakness tags per subject
<code>topic_mastery_json</code>	JSONB	lightweight map <code>{topic: score}</code>
<code>last_active_at</code>	TIMESTAMP	
<code>attempts_total</code>	INT	
<code>hints_used_total</code>	INT	
<code>full_solutions_opened_total</code>	INT	
<code>avg_grader_confidence</code>	NUMERIC(4,3)	
<code>streak_days</code>	INT	optional gamification
<code>updated_at</code>	TIMESTAMP	

This table should remain **small and denormalized**, because it powers the dashboard. Detailed event data belongs elsewhere.

Error logs table

Column	Type	Notes
<code>id</code>	UUID PK	
<code>user_id</code>	UUID FK	
<code>question_id</code>	UUID FK	

Column	Type	Notes
<code>subject_code</code>	VARCHAR(4)	
<code>error_family</code>	TEXT	<code>concept</code> , <code>method</code> , <code>algebra</code> , <code>sign</code> , <code>units</code> , <code>diagram_reading</code> , <code>graph_reading</code> , <code>essay_structure</code> , <code>translation</code> , <code>carelessness</code>
<code>error_atom</code>	TEXT	Specific tag such as <code>forgot_moles_ratio</code> , <code>used_g_not_gsinθ</code>
<code>severity</code>	SMALLINT	1-5
<code>detected_from</code>	TEXT	<code>grader</code> , <code>ocr</code> , <code>user_self_report</code> , <code>reviewer</code>
<code>evidence_json</code>	JSONB	snippets, region refs, rubric refs
<code>resolved</code>	BOOL	for future notebook workflows
<code>created_at</code>	TIMESTAMP	

This table is the most important “futureproofing” decision in the whole MVP. Even if you do not ship a wrong-question notebook UI yet, you should **log the mistakes now**.

API stack and budget model

Recommended provider selection

A single-provider stack is best for production MVP stability, but a two-provider strategy is best for pilot flexibility.

Provider	What to use it for	Pros	Cons	Recommendation	Source
Alibaba Cloud Model Studio 百炼	Core text tutor, OCR, multimodal fallback, math grading	China-mainland deployment, OpenAI-compatible APIs, batch inference, context caching, OCR, Qwen open-model families, DeepSeek vendor routes, versioned model catalog	Free quotas are time-limited; some “latest” aliases can drift; not every cheapest model is the best grader	Default production provider	19

Provider	What to use it for	Pros	Cons	Recommendation	Source
Tencent OCR	Question split, formula OCR, pilot benchmarking	Education-oriented OCR; question recognition supports printed text, handwriting, formulas, and image coordinates; math/formula OCR returns LaTeX; useful free packs	Paid question grading and splitting can exceed budget fast; K12-oriented grading product is not ideal as your core engine	Optional pilot fallback only	20
SiliconFlow	Dev/test backup, A/B testing, occasional fallback	Large catalog of open-source APIs, official OpenAI-compatible quick start, some free models noted by official materials	Promotional/free availability is not a safe production assumption; vendor/rate-limit behavior may change	Use for experimentation, not stable grading core	21

Recommended model and OCR stack

Task	Default choice	Why
Normal chat tutoring	qwen2.5-7b-instruct or qwen3-8b	Cheap, open model family, sufficient for hinting and step explanations
Math-heavy grading	qwen2.5-math-7b-instruct	More reliable for math-specific reasoning and symbolic checking
OCR	qwen-vl-ocr	Extremely cheap token-based OCR, China-mainland deployment, free intro quota
Hard visual fallback	qwen3-vl-8b-instruct	For graphs, diagrams, tables, low-confidence OCR
Exceptional hard visual cases	qwen3-vl-30b-a3b-instruct	Reserve-only route for difficult diagrams/pages
Dev fallback	SiliconFlow free/open model routes	Useful for testing and prompt iteration

This stack keeps the default path cheap and pushes expensive multimodal reasoning into a rare fallback lane. The open-source status of the underlying model families is supported by the official Qwen, DeepSeek, and MiniCPM repositories. ²²

Cost estimates

The most useful answer is not “what is the official price sheet,” but “what will one student action roughly cost if prompts are disciplined?” The table below uses official China-mainland unit prices and conservative request shapes.

Operation	Assumed tokens / calls	Unit pricing basis	Approx cost
Hint-only tutoring	1,500 input / 250 output on <code>qwen2.5-7b-instruct</code>	¥0.5/M input, ¥1/M output	~¥0.0010
Full worked solution	2,200 input / 700 output on <code>qwen2.5-7b-instruct</code>	same	~¥0.0018
Math grading pass	1,800 input / 500 output on <code>qwen2.5-math-7b-instruct</code>	¥1/M input, ¥2/M output	~¥0.0028
OCR parse of 1600×1200 image	about 2,027 input tokens incl. prompt / 800 output on <code>qwen-vl-ocr</code>	¥0.3/M input, ¥0.5/M output	~¥0.0010
Visual fallback	about 2,277 input / 400 output on <code>qwen3-vl-8b-instruct</code>	¥0.5/M input, ¥2/M output	~¥0.0019
Hard visual fallback	same token shape on <code>qwen3-vl-30b-a3b-instruct</code>	¥0.75/M input, ¥3/M output	~¥0.0029

These estimates are derived from Alibaba’s official unit prices and image-token accounting rule for vision inputs. Alibaba documents that image token count scales with image resolution, and its OCR docs point to token-based billing for images. ²³

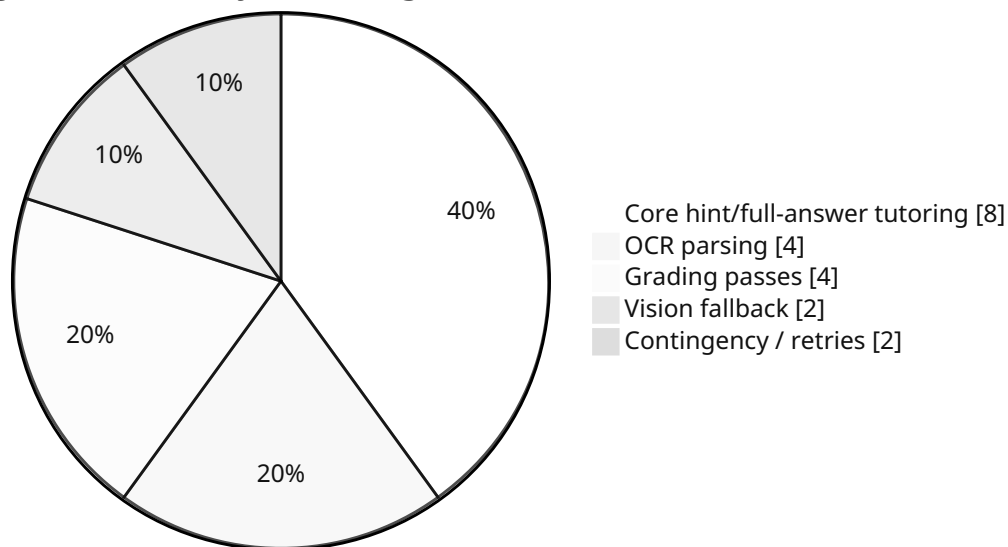
A practical takeaway follows from the math. A **carefully budgeted end-to-end image-heavy review**—OCR parse, hint, optional full solution, grading, and one visual fallback—can still remain in the **sub-¥0.01** range if prompts are short. In real life, however, retries, messy images, larger outputs, and developer experimentation will inflate that. I would therefore plan around a **5×–10× safety factor**, which means a ¥20 API budget should be treated as enough for roughly **250–1,000 meaningful image-to-feedback question reviews per month**, or a few thousand lightweight text-only turns, not more.

Why Tencent paid grading is not the default

Tencent’s education OCR is powerful, but its priced grading pipeline is hard to justify under a ¥20 ceiling. Official Tencent pricing shows **试卷切题 at ¥0.24 per call** and **试题批改 Agent at ¥0.15 per question**. A single 10-question worksheet would therefore cost roughly **¥1.74** before retries or essay handling—about eleven such sheets for the full monthly budget. That is why Tencent should be used as **pilot tooling, QA benchmarking, or fallback**, not the default production grader for this budget. ²⁴

Suggested budget allocation

Suggested monthly API budget inside ¥20



This split only works if you deliberately avoid auto-spending. The product should **not** automatically run full solution generation, grading, and multimodal fallback on every action.

Cost-saving strategies and tradeoffs

The best cost lever is **staging**. Do not generate the worked solution until the student asks. The first response should be a short hint. This is simultaneously better pedagogy and lower spend. ²⁵

The second cost lever is **stable prompt caching**. Alibaba's explicit cache is useful here because your system prompt, syllabus framing, grading rubric, and answer template are mostly fixed. Cache those blocks, and keep only the student-specific delta uncached. Alibaba's documentation explicitly says explicit cache is for stable repeat reuse and can materially reduce repeated-cost scenarios. ²⁶

The third cost lever is **batch, not live, for background intelligence**. Wrong-question notebook generation, nightly topic summaries, and teacher-facing analytics should use batch inference, because Alibaba's batch jobs are officially priced at half the cost of live inference for suitable tasks. ²⁷

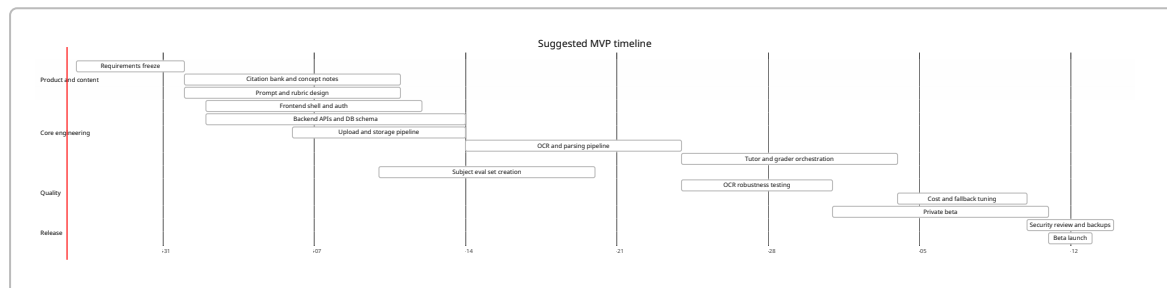
The fourth cost lever is **classical CV before model calls**. Deskew, crop, and compress images before OCR. Even small resolution reductions matter because image token usage scales with pixel count. Alibaba's vision docs provide an image-token formula, so image size directly affects cost. ²⁸

The fifth cost lever is **avoid "latest" model aliases for grading-critical paths**. Pin a snapshot model id in production so rubric behavior does not drift across vendor-side updates. Alibaba explicitly documents model updates and snapshot releases, which is a good reason to version-lock the grading path. ²⁹

Delivery, testing, security, and maintenance

MVP delivery plan

A realistic MVP can be delivered in about **six to eight weeks** if the scope is kept tight.



Milestones

Milestone one should deliver the usable skeleton: sign-in, uploads, text tutoring, and a single subject working end-to-end. Start with **9709 Mathematics**, because it gives the clearest signal on OCR quality, shown-working grading, and stepwise explanations without the added complexity of essay prose or chemistry drawings.

Milestone two should add image parsing, citation rendering, and low-confidence handling. The key success metric here is not “AI impressiveness.” It is whether the system gracefully says **“I cannot read this clearly enough”** rather than inventing certainty.

Milestone three should add homework batch correction for all four subjects, plus the logging needed for the future wrong-question notebook and user profiling.

Milestone four should focus on calibration: prompt trimming, model routing, version pinning, and budget envelopes.

Testing plan

The testing strategy should combine **academic accuracy**, **OCR robustness**, **UX clarity**, and **cost regression**.

Test stream	What to test	Minimum MVP target
Unit tests	Prompt assembly, routing logic, schema validation, rubric calculators	80%+ coverage on orchestration code
OCR tests	Clear print, low light, skew, handwritten workings, graphs, mixed Chinese/English notes	At least 30 image cases per subject
Subject eval set	Gold questions with expected hints, full solutions, and misconception tags	At least 50 items per subject
Grading tests	Exact answer, near miss, method-only credit, unit errors, essay structure	4–6 rubric cases per question type

Test stream	What to test	Minimum MVP target
Citation tests	Every substantive answer block must reference retrieved source ids	100% pass gate in production
Safety tests	Prompt injection in uploads, malicious filenames, oversized PDFs, image bombs	Block or isolate all
Cost tests	Per-route token usage and fallback frequency	Daily budget alarms and route caps
UX tests	EN-first / ZH-first / exam-English mode, hint-to-solution flow	Task completion with students in pilot

The evaluation set should be built from **teacher-authored items, user-uploaded homework, exemplar questions, and school-permitted materials**, with careful rights handling. Cambridge's official policies are a good reason not to build the MVP around redistributing a public answer bank. ³⁰

Security and privacy

The privacy baseline should be stronger than a normal study app because education data often includes minors, handwritten work, and performance histories. Under China's PIPL, under-14 data is sensitive personal information and requires dedicated handling rules; the minors' network protection rules also emphasize minimum necessary access, staff access controls, and event logging. ³¹

For the MVP, the safest operational policy is:

- default to **China-mainland compute and storage**;
- use **signed upload URLs** and strip EXIF where possible;
- encrypt storage at rest and TLS in transit;
- keep **PII separate from learning telemetry**;
- support account deletion and export;
- add a clear **guardian-consent path** if you ever admit under-14 users;
- avoid storing original files longer than necessary when the normalized question JSON is sufficient.

That recommendation is strengthened by the cross-border rules: if you send personal information to providers or infrastructure outside China, you may trigger additional transfer obligations. A China-region deployment avoids that complexity in the MVP phase. ³²

Because the product is a generative AI service, you should also treat content governance seriously. China's Interim Measures for Generative AI Services apply to public-facing generative services in China. In practice, for this MVP, that means you should add content filters, abuse reporting, low-confidence disclaimers, audit logs, and a clear way to escalate failures. Using established domestic providers for the model layer simplifies that compliance posture. ³³

Maintenance plan

The most important maintenance rule is **version everything**:

- prompt version,
- rubric version,

- model snapshot id,
- citation-bank version,
- OCR routing thresholds.

Without this, you will not be able to debug grading regressions.

The second rule is **monitor cost and confidence together**. A low-cost route that raises low-confidence flags too often is not actually cheaper, because it causes rework and fallback calls. Track:

- OCR confidence,
- fallback rate,
- average tokens per route,
- hint-to-full-solution conversion,
- grading disagreement rate on review samples.

The third rule is **human QA on high-risk categories**. In this product, those categories are:

- messy handwritten math,
- complex chemistry structures/mechanisms,
- long economics essays,
- ambiguous graphs and practical diagrams.

The fourth rule is **grow the notebook and profile features from logs, not from guesswork**. If you record clean error atoms from day one, your later “wrong-question notebook” can group mistakes by topic, misconception, and language mode with no redesign.

Final recommended MVP stack

If I had to choose a single concrete stack for immediate implementation, it would be:

- **Frontend:** Next.js + TypeScript + Tailwind + shadcn/ui
- **Backend:** FastAPI + Pydantic + SQLAlchemy/SQLModel
- **Storage:** PostgreSQL + Redis + Alibaba OSS
- **Auth:** email OTP or password login in FastAPI for MVP; add phone login later
- **Core APIs:**
 - `qwen2.5-7b-instruct` or `qwen3-8b` for tutoring
 - `qwen2.5-math-7b-instruct` for symbolic/math grading
 - `qwen-vl-ocr` for OCR
 - `qwen3-vl-8b-instruct` for graph/diagram/image fallback
- Tencent OCR free packs only as optional pilot helper
- **Deployment:** China-region Docker deployment on one small VM plus OSS backup for pilot; move DB to managed service once usage stabilizes

That stack best satisfies the user constraints: **web-only MVP, no local models, China-hosted open-source APIs, bilingual tutoring, image-based recognition, explainability, and very tight API cost control**. The critical product discipline is not the model choice by itself. It is the routing policy: **cheap hint first, full answer only on demand, OCR before vision, grading after normalization, and citations in every substantive answer**.

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